Basics of derivatives contracts

William Mann

EMORY
GOIZUETA
BUSINESS
SCHOOL

Overview

- Derivatives are contracts that specify payments which depend on (*derive* from) the price of some other asset.
- Like a short sale, they are:
    - zero-sum arrangements between investors or other market participants,
    - can change the basic pattern of payoffs,
    - can create leverage by amplifying payoffs relative to invested capital.
- They can be used to transfer/manage risk, and to express speculative bets.

Today’s outline:

- Over-the-counter (OTC) derivatives: Forwards, swaps, and options.
- Exchange-traded derivatives: Futures and options.
- Examples and applications.
- Basic economic principles.

We will *not* go in-depth with options pricing.

OTC derivatives contracts

- Over-the-counter (OTC) derivatives are contracts between you and a dealer.
- The dealer doesn't necessarily take the other side of your contract. They deal with many people on both sides of the market, and hedge their net exposure across these positions by trading in the underlying asset or other derivatives.
- You first must establish an ISDA Master Agreement with the dealer, which will have standard provisions related to contract terms, collateral, and other details. Then you can enter into contracts fairly simply by communicating directly with the dealer.
- While you can imagine a wide range of contracts being signed, a few varietes are extremely popular and will be our focus: Forwards, swaps, and options.

Forwards

- An agreement today on a price called the *forward price* at which we will trade an asset in the future.
- Extremely useful for locking in prices and removing risk, for example related to commodity prices, exchange rates, and interest rates.
- The forward price is not necessarily equal to the *spot* price, which refers to the price you would pay on the market to buy the asset right now. In fact, there will be a simple no-arbitrage relation between the spot price, interest rates, and costs or benefits of carry. See later.
- We will also see later that futures contracts have a very similar structure. The difference is where they are avilable: By definition, forward contracts are OTC, and futures contracts are exchange-traded.

Swaps

- An agreement to literally swap two streams of payments. Common examples:
    - Interest rate: fixed for floating rates.
    - Currency swap: loan payments in different currencies.
    - Commodity swap: effectively a series of forward contracts.
- In fact, every swap can be seen as a series of forward contracts.
- Most swaps are set up so that there is zero present value of its future transfers, valued by forward-contract pricing. That way no money needs to change hands up front.
- Swaps are inherently customized and complex, so they are only available OTC.

Options

- The right, but not the obligation, to buy or sell an asset in the future at a set price called the *exercise price* or *strike price*.
    - Call option: Gives the contract owner the right to *buy* at the exercise price.
    - Put option: Gives the contract owner the right to *sell* at the exercise price.
    - Of course the contract owner must pay the counterparty up front to take the other side of this situation.
- There is a set maturity date after which the option is worthless if not exercised.
    - *American-style* options can be exercised at any time up to that date.
    - *European-style* options can only be exercised at the expiration date.
    - These terms are for historical reasons only and are no longer connected to literal geography.

Exchange-traded derivatives contracts

- Some categories of contracts are so popular on both sides of the market that they have become standardized and listed on exchanges, where most trading volume now happens.
- In this setting, you trade with other investors, but with the exchange's clearinghouse between you as a counterparty. It provides guarantees that eliminate counterparty risk, and sets strict margining requirements to protect itself.
- **Options** are fairly similar in the exchange-traded setting as OTC, but with standardized strike prices, expiration dates, and contract amounts.
- **Futures** is the name for an exchange-traded forward contract. They are the same basic structure, but subject to many strict rules. Most importantly, margin requirements are marked to market *and settled* every day.
- **Swaps** are only available OTC because they are inherently customized and complex.

Use cases

Derivatives are useful in a huge range of settings both for risk management, and for constructing strategies with a desired payoff structure. Examples from this class:

- May commodity ETFs hold futures contracts instead of the physical commodity.
- Leverage and inverse ETFs use futures or swaps. More generally, many funds use derivatives instead of borrowing or short-selling directly to achieve leverage or short exposure.
- Treasury basis trade (cash vs futures) is a major strategy in fixed income.

Measuring the size of derivatives markets

- How big are derivatives markets?
- The ``notional value'' of the market adds up the reference amount of all contracts, without netting long and short sides. This is typically in the hundreds of trillion dollars.
- This vastly overstates the money at stake, since participants are usually trading much less than the principal value and may take offsetting positions with each other.
- On the other hand, long and short positions have to offset exactly to zero in aggregate, so in a macroeconomic sense the size of the market is always zero, yet this is also a misleading perspective.
- Regulators pay more attention to numbers in between these two extremes. For example:
    - ``Gross market value'' is the sum of the market value of each contract to the side that has positive value. This is typically less than 5% of notional value.
    - ``Gross credit exposure'' is the above concept if we also net out offsetting positions between counterparties. This is typically less than 1% of notional value.

Pricing of forwards, futures, and swaps

For an underlying asset that pays no dividends and can be stored at no cost, then no-arbitrage logic implies a surprisingly simple connection between the spot price S and the forward price F:

$$F = S \times (1 + r_f)^T$$

If the asset pays dividends, the forward price will be lower; if it is costly to store the forward price will be higher.

The surprising thing is that the asset's expected return does not appear in this formula. The justification for this comes from an arbitrage argument: If the price were anything other than the above relationship, one could easily buy the cheaper side, sell the more expensive side, and borrow or save at the risk-free rate to form a risk-free profit.

Swaps are best modeled as a stream of forward contracts. Each future payment can be valued using forward pricing formulas and their total amounts added together. The future payment terms will be set so that each side gets a fair deal and no money needs to change hands today.

Options exercise decisions and payoffs

- As the option contract approaches expiration, the contract owner it will exercise when that is optimal, and otherwise will let the option expire with no cash flow.
- For a call option, the owner exercises if the asset's current market price is **above** the exercise price.
    - In this case they buy the asset from their counterparty at the exercise price, sell into the market for the market price.
- For a put option, the owner exercises if the asset's current market price is **below** the exercise price.
    - In this case they buy the asset from the market at the market price, sell it to their counterparty for the exercise price.
- Some options are cash-settled, meaning the seller of the contract just pays the difference between market price and exercise price.

Economic insights about options

Based on the earlier logic we can reason that:

- For **put options:** The potential benefit from buying them, and the potential loss from selling them, are both limited to the exercise price.
- For **call options:** The potential benefit from buying them, and the potential loss from selling them, are both *unlimited*.
  - This is like taking a short position. Margin requirements become a major issue with these positions.
- **All options** are more valuable when their underlying asset becomes more volatile.
  - You only exercise them when advantageous, and volatility raises the chance of that happening.
- Put-call parity: Long call + cash = Long stock + long put.
  - Suppose you buy a call option with exercise price K, and save enough at the risk-free rate to have K dollars at expiration.
  - This has exactly the same payoff as if you buy the underlying asset and a put option with the same exercise price.
  - No-arbitrage suggests that the price of the call option plus $K/(1+r_f)^T$ should equal the price of the underlying asset plus the price of the put option.

Embedded derivatives and real options

Even when you are not trading literal derivatives, insights from these markets can influence your decision making in other settings.

Many contracts are best understood as having ``embedded'' derivatives, for example:

- Convertible and callable securities offer one side of the deal a valuable decision.
- Corporate default swaps (CDS) generate one-time contingent insurance-like payoffs.
- For a firm with large debt levels, equity itself resembles a call option, while debt is like a portfolio with a short position in a put option. This insight is behind ``structural'' models of corporate credit risk.

Many important decisions can be thought of as ``real options,'' such as developing new oil reserves, movie sequels, or medicines.

- These opportunities can be valued with options-based techniques that go beyond simpler NPV analysis and reveal insights based on options pricing.